

Lecture 03 : Basic Concepts of Networking (part-II)

Introduction

In the intricate world of computer networking, understanding how data traverses vast distances is fundamental. This lecture delves into the second part of basic networking concepts, building upon previous discussions of data transmission approaches. We begin by exploring the datagram approach, a flexible and widely adopted method for packet switching, contrasting it with the virtual circuit model. Following this, we unravel the complexities of network design through the lens of layered network architecture, specifically the Open Systems Interconnection (OSI) model. This conceptual framework is crucial for designing robust, maintainable, and scalable networking systems, providing a structured way to understand the various functions involved in data communication. By the end of this session, students will grasp the core principles governing data transmission and the architectural paradigms that underpin modern networks.

Core Concepts

Term	Definition	Significance
Datagram Approach	A packet switching method where each packet is treated as an independent entity, routed individually without prior path establishment.	Offers flexibility and resilience, widely used in the internet due to its adaptability to network changes.
OSI Model	A conceptual framework that standardizes the functions of a telecommunication or computing system into seven distinct layers.	Provides a universal standard for network design, promoting interoperability and modular development of network components.
Propagation Delay	The time it takes for a signal to travel from the source to the destination across a physical medium.	A critical factor in network performance, influenced by the physical characteristics of the transmission medium (e.g., fiber optic, satellite).

Term	Definition	Significance
Transmission Time	The time required to push all bits of a message onto the transmission medium.	Depends on the message size and the bandwidth of the link, distinct from propagation delay.
Network Layer	The third layer of the OSI model, responsible for packet routing, addressing, and forwarding across multiple network segments.	Essential for inter-network communication, ensuring packets reach their correct destination across diverse networks.
Transport Layer	The fourth layer of the OSI model, providing end-to-end reliable data transfer between applications on different hosts.	Guarantees data integrity and delivery, managing connection establishment, maintenance, and termination for applications.
Router	A network device that forwards data packets between computer networks, operating at the network layer (Layer 3).	Connects different networks, making intelligent forwarding decisions based on IP addresses and routing tables.

Detailed Analysis

1. The Datagram Approach to Packet Switching

The datagram approach stands as a primary alternative to the virtual circuit approach in packet switching. Its fundamental characteristic is the absence of a pre-established route. Unlike virtual circuits, where a dedicated path is set up before data transmission, each packet in the datagram approach is treated as an independent entity. This is analogous to the postal system, where each letter is dispatched individually, carrying its destination address, without a predefined route for all letters from a single sender to a single receiver.

Key aspects of the datagram approach include:

- **No Route Establishment:** There is no initial delay associated with setting up a connection path, which can make it faster for transmitting a small number of packets.
- **Independent Packet Transmission:** Each packet is routed independently, meaning they may follow different paths to the same destination. This requires each

packet to carry the destination address in its header.

- **Dynamic Routing Decisions:** Intermediate nodes maintain routing tables and make dynamic decisions on where to forward each incoming packet. This allows for flexibility in adapting to network conditions.

- **Potential Disadvantages:**

- **Out-of-Order Delivery:** Since packets can take different paths, they may arrive at the destination in an order different from their original transmission sequence.

- **Packet Loss:** If an intermediate node temporarily fails or its buffer overflows, packets stored there can be lost.

- **Duplicate Packets:** In scenarios where acknowledgements are delayed or lost, a sender might retransmit a packet, leading to duplicates at the receiver.

- **Advantages:**

- **Flexibility and Resilience:** The ability for packets to follow different paths allows the network to handle congestion or link failures by rerouting traffic.

- **No Connection Overhead:** Eliminates the initial setup and termination overheads, making it efficient for bursty traffic or short communications.

- **Congestion Management:** Can adapt to network load by distributing packets across less congested paths.

While the datagram approach offers significant flexibility and is widely used in the internet, its efficiency compared to virtual circuits depends on the volume of data. For a large number of packets, the initial setup cost of a virtual circuit might be offset by its faster, dedicated transmission phase.

2. Components of Network Delay

Understanding network performance requires analyzing the various components that contribute to overall delay in data communication:

- **Propagation Delay:** This is the time a signal takes to travel from the source to the destination. It is solely dependent on the physical medium and the distance. For instance, a satellite link will have a significantly higher propagation delay than a copper or fiber optic link due to the longer distance signals must travel.

- **Transmission Time:** This refers to the time required to transmit all the bits of a message onto the communication link. It is determined by the size of the message (number of bits) and the bandwidth (speed) of the link. For example, sending 1000 bits over a 1 Mbps link would take 1 millisecond ($1000 \text{ bits} / 1,000,000 \text{ bits/second} = 0.001 \text{ seconds}$). Transmission time is independent of the physical medium's propagation characteristics.

- **Processing Delay:** When data passes through intermediate nodes (like routers), these nodes need time to receive the packet, store it (store and forward), consult their routing tables, make forwarding decisions, and then transmit the packet. This small but cumulative delay at each intermediate node is the processing delay.

These three components collectively determine the total delay experienced by data packets in a network.

3. Comparison of Switching Approaches: Circuit, Virtual Circuit, and Datagram

The lecture highlights three primary approaches to data switching:

- **Circuit Switching:** Involves an initial circuit establishment phase, after which a dedicated, fixed-bandwidth connection is maintained between source and destination. Once established, data transmission occurs at maximum speed continuously without further delay, as the bandwidth is guaranteed.
- **Virtual Circuit Packet Switching:** Begins with a "call request packet" to establish a virtual circuit, setting up routing tables in intermediate nodes. Once the circuit is established and confirmed by a "call accept packet," data packets flow sequentially in a pipelined fashion, using a store-and-forward approach. While it has an initial setup delay, subsequent packet transmission is efficient as the path is known.
- **Datagram Packet Switching:** As discussed, this approach has no connection establishment phase, meaning no initial delay. Packets are sent independently, potentially following different paths, and also utilize the store-and-forward concept. Its strength lies in flexibility and resilience to network changes, making it the dominant approach in the internet, despite potential for out-of-order delivery or higher total time for very large data transfers compared to a well-established virtual circuit.

4. Layered Network Architecture: The OSI Model

To manage the complexity of network design, a conceptual layering approach is adopted, dividing the total functionality into well-defined levels. The Open Systems Interconnection (OSI) model, proposed by the International Organization for Standardization (ISO), is a prominent example, comprising seven layers.

The advantages of layering include:

- **Systematic Design:** Breaks down complex networking tasks into smaller, manageable functions.
- **Modularity and Maintainability:** Allows modifications or technological upgrades to one layer without affecting others.
- **Interoperability:** Provides a standard framework for different systems to communicate.

The OSI model's layers are:

1. **Physical Layer**
2. **Data Link Layer**

3. **Network Layer**
4. **Transport Layer**
5. **Session Layer**
6. **Presentation Layer**
7. **Application Layer**

A crucial distinction is made between **host-to-host** layers (upper 4: Transport, Session, Presentation, Application) and **point-to-point** layers (lower 3: Physical, Data Link, Network). Host-to-host layers are active only at the source and final destination, while point-to-point layers are active at all intermediate nodes along the path. Data flows down the stack at the source, across the physical medium, and up the stack at the destination. Intermediate nodes process data up to the network layer (or data link/physical for specific devices) before forwarding it.

5. Functions of OSI Layers

Each layer in the OSI model performs specific functions:

- **Physical Layer (Layer 1):** Deals with the physical transmission of raw bits over a communication channel. This includes electrical signals, optical signals, radio waves, and the physical medium itself (e.g., copper cable, fiber optic, microwave link, Bluetooth). It establishes a physical connection between directly connected points.
- **Data Link Layer (Layer 2):** Ensures reliable transfer of data (called "frames") over a point-to-point link. It handles error detection and retransmission, flow control (managing data transfer rates to prevent speed mismatches), and provides a reliable connection between two directly connected nodes.
- **Network Layer (Layer 3):** Responsible for packet routing across multiple interconnected networks. It establishes, maintains, and terminates connections, and most importantly, forwards packets in the correct direction through potentially many intermediate computers to reach the ultimate destination.
- **Transport Layer (Layer 4):** Provides end-to-end reliable data transfer between applications. It ensures that data sent from a source application arrives correctly at the destination application, handling retransmissions if necessary, without concern for intermediate nodes.
- **Session Layer (Layer 5):** Manages communication sessions between applications. This includes establishing, coordinating, and terminating conversations, such as managing login/logout processes in an application.
- **Presentation Layer (Layer 6):** An optional layer that handles data representation and syntax. It can perform code conversion (e.g., ASCII to EBCDIC), data encryption/decryption, and compression/decompression, ensuring that data is presented in a format understandable by the application layer.
- **Application Layer (Layer 7):** The topmost layer, providing network services directly to end-user applications. This includes protocols for email (SMTP), web

browsing (HTTP), file transfer (FTP), and any other application that uses the network for data transmission.

Data flow through this "network stack" involves encapsulation as it moves down from the application layer to the physical layer at the source, and decapsulation as it moves up from the physical layer to the application layer at the destination.

6. Internetworking Devices

Various devices operate at different layers of the OSI model to facilitate network communication:

- **Hubs:** Operate at the Physical Layer (Layer 1). They simply broadcast incoming data to all connected devices, without any intelligence regarding destination.
- **Bridges / Layer-2 Switches:** Operate at the Data Link Layer (Layer 2). They learn the MAC addresses of devices on connected segments and forward data only to the intended segment, reducing network traffic.
- **Routers / Layer-3 Switches:** Operate at the Network Layer (Layer 3). They connect different networks and make intelligent forwarding decisions based on IP addresses and routing tables. Routers are typically used to connect an organization's network to the outside world (e.g., the internet). Layer-3 switches perform similar routing functions but are often optimized for internal network traffic within an organization.

A typical network infrastructure often features a hierarchical arrangement:

- **Routers** connect the internal network to external networks.
- **Layer-3 Switches** act as core switches within an organization, handling high-speed routing between different internal network segments.
- **Layer-2 Switches** are used at the departmental or workgroup level to connect end devices.
- **Hubs** (though less common in modern networks) might be found at the lowest level, connecting individual computers.

This layered approach to devices mirrors the layered architecture of network protocols, ensuring efficient and scalable network design.

Key Takeaways

- [] The datagram approach routes each packet independently, offering flexibility and resilience without prior path establishment.
- [] Datagrams can lead to out-of-order delivery, packet loss, and duplicate packets due to independent routing.

- [] Network delays comprise propagation delay (medium-dependent signal travel time), transmission time (message size and bandwidth dependent), and processing delay (time at intermediate nodes).
- [] The OSI model divides network functionality into seven distinct layers to simplify design, improve maintainability, and ensure interoperability.
- [] The lower three OSI layers (Physical, Data Link, Network) are active at intermediate nodes, while the upper four (Transport, Session, Presentation, Application) are end-to-end.
- [] Each OSI layer has a specific function, from raw bit transmission (Physical) to end-user application services (Application).
- [] Different internetworking devices (hubs, switches, routers) operate at specific OSI layers, performing functions appropriate to that layer.
- [] The datagram approach is the most widely used method in the internet due to its flexibility and ability to handle dynamic network conditions.

Conclusion

This lecture has provided a comprehensive overview of the datagram approach to packet switching, highlighting its operational mechanics, advantages, and potential challenges. We've also dissected the critical components of network delay, offering a clearer picture of factors influencing data transmission speed. Crucially, the session introduced the layered network architecture, exemplified by the OSI model, as a fundamental paradigm for designing and understanding complex networking systems. By breaking down network functions into distinct layers, the OSI model facilitates systematic development, modularity, and interoperability, which are cornerstones of modern communication. The discussion on internetworking devices further illustrated how these architectural principles translate into tangible hardware, forming the backbone of network infrastructure. This foundational knowledge sets the stage for our next exploration into specific network protocol stacks, such as TCP/IP, which is the very standard upon which the entire internet operates.